Mini Project 3 - Country Comparison

DAX MEASURE DOCUMENTATION & DATA DICTIONARY

Repository: DAX Measures Table
Total Measures: 26 Calculated Metrics
Status: Validated & Production-Ready

Document Overview

This document details the complete DAX measures repository created for the **Mini Project 3 - Country Comparison** Power BI report. It outlines business logic, calculations, and analytical purpose categorized according to the data model's fact tables.

1. DEMOGRAPHIC HEALTH MEASURES

MEASURE NAME	DESCRIPTION & BUSINESS LOGIC	DAX EXPRESSION
Avg Birth Rate	Calculates the mean birth rate per 1,000 population across selected countries or regions.	`Avg Birth Rate = AVERAGE('Fact Demographic Health'[Birth rate(births/1000 population)])`
Avg Death Rate	Computes the average mortality rate per 1,000 population across filtered geographical subsets.	`Avg Death Rate = AVERAGE('Fact Demographic Health'[Death rate(deaths/1000 population)])`
Avg Infant Mortality	Measures average infant death rate per 1,000 live births to assess baseline healthcare quality.	`Avg Infant Mortality = AVERAGE('Fact Demographic Health'[Infant mortality rate])`
Avg Life Expectancy	Determines average population life expectancy at birth in years across evaluated areas.	`Avg Life Expectancy = AVERAGE('Fact Demographic Health'[Life expectancy at birth(years)])`
Total Population	Aggregates cumulative population across all context-filtered nations.	`Total Population = SUM('Fact Demographic Health'[Population])`
Unemployment Rate	Calculates the average national unemployment percentage across selected territories.	`Unemployment Rate = AVERAGE('Fact Demographic Health'[Unemployment rate(%)])`

2. ECONOMIC METRICS MEASURES

MEASURE NAME	DESCRIPTION & BUSINESS LOGIC	DAX EXPRESSION
Avg GDP	Calculates average nominal Gross Domestic Product (GDP) across countries.	`Avg GDP = AVERAGE('Fact Economic Metrics'[GDP])`
Avg GDP Growth	Computes the mean annual real GDP growth percentage to evaluate economic momentum.	`Avg GDP Growth = AVERAGE('Fact Economic Metrics'[GDP - real growth rate(%)])`
Avg GDP per Capita	Determines mean GDP per capita to reflect	

3. ENERGY RESOURCES MEASURES

MEASURE NAME	DESCRIPTION & BUSINESS LOGIC	DAX EXPRESSION
Countries with Oil Reserves	Counts distinct count of countries possessing verified natural oil reserves.	`Countries with Oil Reserves = CALCULATE(DISTINCTCOUNT('Dim Country'[Country]), 'Fact Energy Resources'[Oil - proved reserves(bbl)] > 0)`
Total Electricity Consumption	Sums national electricity consumption in kilowatt-hours (kWh).	`Total Electricity Consumption = SUM('Fact Energy Resources'[Electricity - consumption(kWh)])`
Total Electricity Generation	Aggregates total electrical energy produced in kilowatt-hours (kWh).	`Total Electricity Generation = SUM('Fact Energy Resources'[Electricity - production(kWh)])`
Total Gas Consumption	Aggregates natural gas consumption volume in cubic meters (cu m).	`Total Gas Consumption = SUM('Fact Energy Resources'[Natural gas - consumption(cu m)])`
Total Gas Production	Sums total natural gas volume produced in cubic meters (cu m).	`Total Gas Production = SUM('Fact Energy Resources'[Natural gas - production(cu m)])`
Total Oil Consumption	Calculates total daily oil consumption in barrels per day (bbl/day).	`Total Oil Consumption = SUM('Fact Energy Resources'[Oil - consumption(bbl/day)])`
Total Oil Production	Calculates total daily oil production in barrels per day (bbl/day).	`Total Oil Production = SUM('Fact Energy Resources'[Oil - production(bbl/day)])`

4. INFRASTRUCTURE MEASURES

MEASURE NAME	DESCRIPTION & BUSINESS LOGIC	DAX EXPRESSION
Highway Length (Km)	Aggregates overall paved and unpaved roadway infrastructure length in kilometers.	`Highway Length (Km) = SUM('Fact Infrastructure'[Highways(km)])`
Railway Length (Km)	Sums total active railway network distance measured in kilometers.	`Railway Length (Km) = SUM('Fact Infrastructure'[Railways(km)])`
Internet Hosts	Aggregates the total number of registered domain internet hosts.	`Internet Hosts = SUM('Fact Infrastructure'[Internet hosts])`
Internet Users	Calculates cumulative internet user count across selected territories to assess digital	`Internet Users = SUM('Fact In`